

Core Algorithms & Data Structures

1. Bloom Filter — Deduplication

- **Problem:** Duplicate records from at-least-once Kafka delivery
- **Solution:** Each record is hashed to a unique key, checked against the Bloom Filter
- **Benefits:**
 - Constant-time membership testing
 - Avoids expensive join operations
 - Highly memory-efficient with minimal accuracy impact
- **Tradeoff:** Small false-positive rate (may occasionally flag a non-duplicate as duplicate)

2. HyperLogLog (HLL) — Cardinality Estimation

- **Purpose:** Estimate number of **distinct flight routes**
- **Memory:** Sub-linear — far less than storing all unique values
- **Use case:** Knowing approximate cardinality at scale without exact computation

3. Count-Min Sketch (CMS) — Frequency Estimation

- **Purpose:** Track **frequency of elements** (tail numbers, delay causes)
- **Use case:** Identifying heavy hitters in streaming data
- **Memory:** Sub-linear with approximate counts

4. MinHash Locality Sensitive Hashing (LSH)

- **Purpose:** Detect routes with **similar delay behavior**
- **How it works:**
 1. Convert route delay features into MinHash signatures
 2. LSH groups similar signatures into buckets
 3. No pairwise comparisons needed (computationally scalable)
- **Insights revealed:** Short-haul vs. long-haul delay pattern clusters
- **Operational value:** Apply targeted mitigation strategies across similar route groups

5. GraphX PageRank — Delay Propagation Hubs

- **Graph model:**
 - Nodes = airports

- Edges = flight routes (weighted by delay metrics)
- **PageRank vs. Degree Centrality:**
 - Degree: counts direct connections
 - PageRank: captures *recursive* importance based on full network structure
- **Key finding:** Some airports have low traffic volume but high propagation influence (e.g., DEN)

6. GraphX Connected Components

- **Purpose:** Identify clusters of closely related airports
- Complements PageRank by revealing subgraph structure

7. Differential Privacy

- **Mechanism:** Laplace noise injection on aggregate statistics
- **Epsilon (ϵ) tradeoff:**
 - Low $\epsilon \rightarrow$ high privacy, high noise
 - High $\epsilon \rightarrow$ low privacy, low noise
 - Balanced $\epsilon \rightarrow$ strong privacy with minimal utility loss
- **Also includes:** Fairness auditing (check for bias across airline categories), data generalization (reduce re-identification risk)

Memory vs. Accuracy Summary

Algorithm	Memory Usage	Accuracy
Bloom Filter	Very low	~100% (small FP rate)
HyperLogLog	Sub-linear	~98%+
Count-Min Sketch	Sub-linear	~98%+
Exact methods	Linear (high)	100%

Probabilistic algorithms reduce memory by **over 99%** vs. exact methods